Algebra 3 — Factorising · Fractions · Equations with fractions · Simultaneous equations

No calculator. Marks in brackets. ★ = harder question. Check every solution in the original equation(s).

1 Factorise completely: [2]

a $5x^2 + 10x$ **b** $18a^2b - 12ab^2$

2 Simplify: [4]

a $\dfrac{x^2 - 3x}{3x}$ **b** $\dfrac{a - ab}{b - b^2}$

3 Solve: [8]

a $\dfrac{2(x-5)}{3} = -4$ **b** $\dfrac{x-2}{4} = \dfrac{x+2}{5}$ **c** $\dfrac{x}{2} - \dfrac{x}{6} = 3$ **d** $\dfrac{6}{x} = -\dfrac{2}{5}$

4 Solve these simultaneous equations. [6]

a $y = 2x - 5$ and $3x + y = 10$

b $3x + 2y = 12$ and $5x - 2y = 4$

5 At a market, 3 oranges and 2 mangoes cost €3.90, while 2 oranges and 5 mangoes cost €5.90. Find the cost of one orange and of one mango. [5]

6 Solve $\dfrac{x+1}{2} - \dfrac{2x-1}{3} = 1$ [3]

7 The sum of two numbers is 27 and their difference is 5. Write two equations and solve them to find the numbers. [4]

Total: 32 marks

Check: clear fractions by multiplying by the lowest common denominator · elimination — match a coefficient, then add or subtract · substitute the pair back into *both* equations.